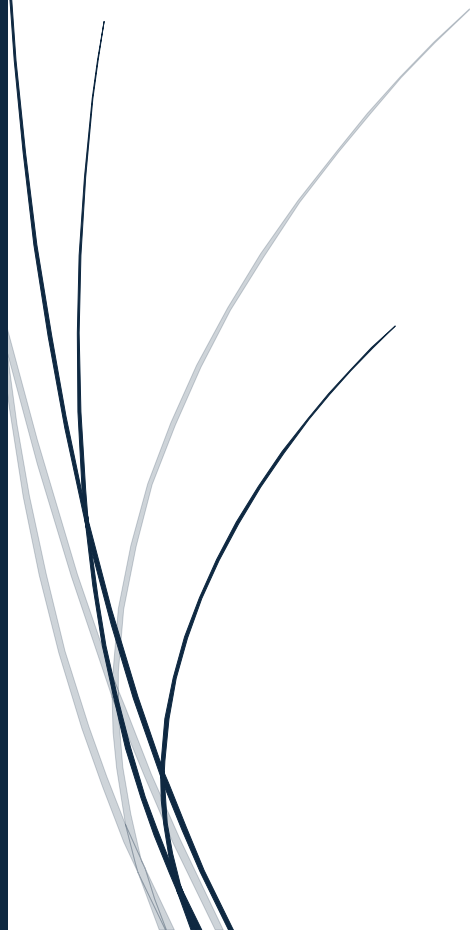


09/05/2026

Insight summary for stakeholders

Dataset Used: Adult Psychiatric Morbidity
Survey (APMS) 2014



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DATA ANALYTICS INTERN

Introduction

During Week 1, an initial exploratory analysis of the Adult Psychiatric Morbidity Survey (APMS-2014) dataset was conducted to better understand the structure, quality, and patterns present within the mental health data. The work focused on data profiling, cleaning, demographic analysis, and regional visualization techniques. The purpose of this analysis was to identify meaningful trends in mental health disorder prevalence while ensuring the dataset was suitable for future analytical and predictive tasks.

1. Dataset Overview

The APMS-2014 dataset contains information related to mental health conditions across different demographic groups. The available data included:

- A. Age groups
- B. Gender groups (Men, Women, and All Adults)
- C. Mental health disorder categories

The dataset was relatively small but structured, containing both numerical and categorical information. Numerical fields represented prevalence percentages across age groups, while categorical variables identified disorder categories and population groups.

2. Data Quality Assessment

An important part of the analysis involved evaluating the quality and reliability of the dataset.

Missing Values

Several numerical columns contained missing information, with approximately 20% of values absent in each age-group column. Missing values can affect the accuracy of statistical findings and future predictive models if not handled carefully.

To improve data reliability:

- A. Missing numerical values were handled using median-based replacement methods.
- B. Categorical fields were checked and found to be complete.

Column	Data Type	Missing Values	Missing %	Unique Values
Category	object	0	0.00%	12
16-24	float64	5	20.83%	19
25-34	float64	5	20.83%	19
35-44	float64	5	20.83%	19
45-54	float64	5	20.83%	19
55-64	float64	5	20.83%	19
65-74	float64	5	20.83%	19
75+	float64	5	20.83%	19
All	float64	5	20.83%	18
Group	object	0	0.00%	3

Duplicate Records

No duplicate entries were identified in the dataset. This indicates that each observation represents a unique record and reduces the risk of overcounting or biased analysis.

Data Formatting

The dataset required minor formatting and normalization steps to ensure that all variables could be analyzed consistently. Numerical fields were standardized and categorical variables were encoded into machine-readable formats for future analytical work.

3. Demographic Breakdown Findings

A demographic analysis was conducted to examine how mental health disorder prevalence varies across age groups and gender categories.

Age-related Trends

The analysis suggests that mental health disorder prevalence differs across age bands. Certain younger and middle-aged groups showed higher variability in prevalence values compared with older age groups.

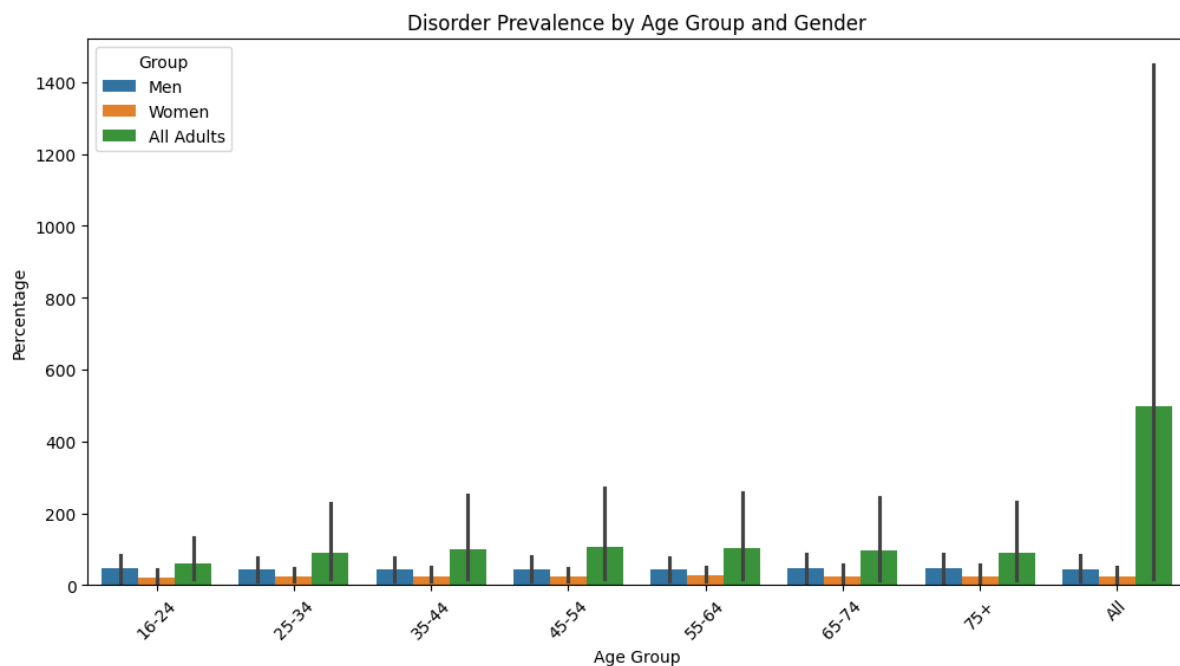
This finding highlights the importance of age-sensitive mental health strategies and targeted interventions.

Gender-related Trends

Differences were observed between men and women across several mental health disorder categories. Some categories showed relatively higher prevalence among women, while others were more evenly distributed.

The “All Adults” category provided a useful overall benchmark for comparison.

These findings reinforce the importance of considering demographic factors when planning mental health services and interventions.



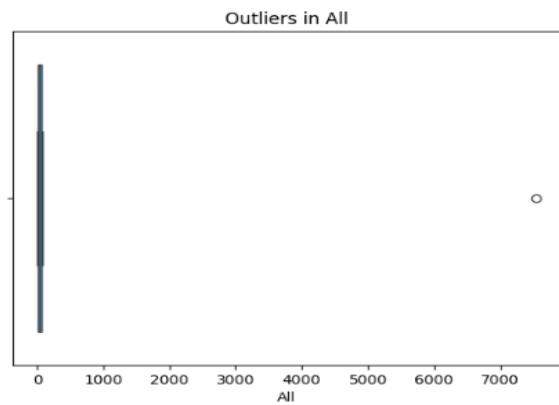
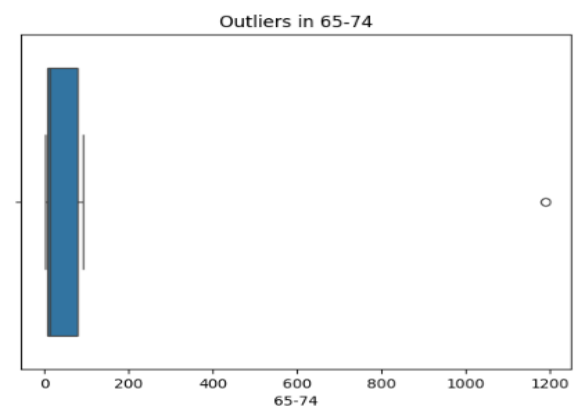
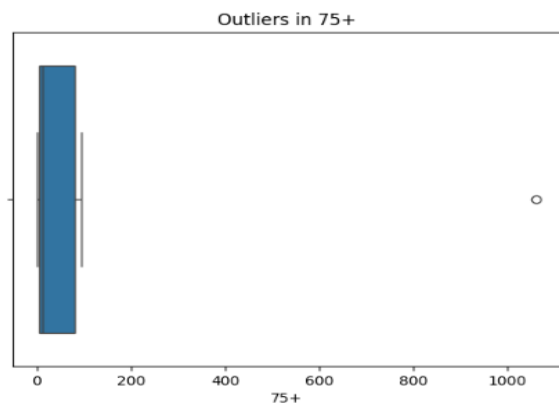
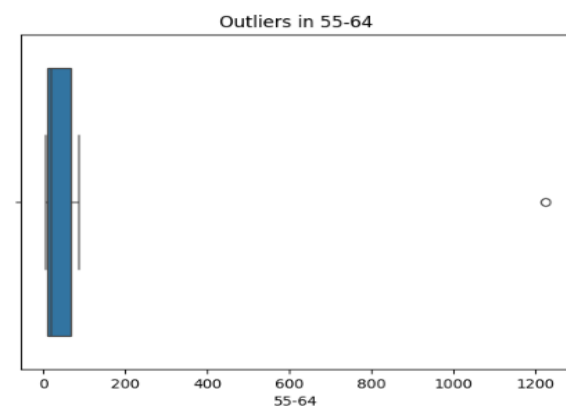
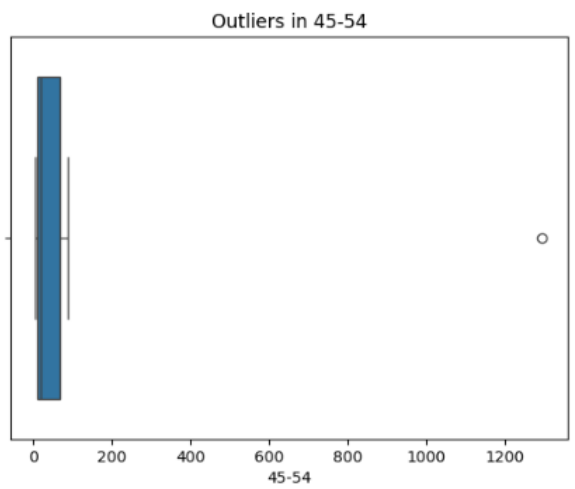
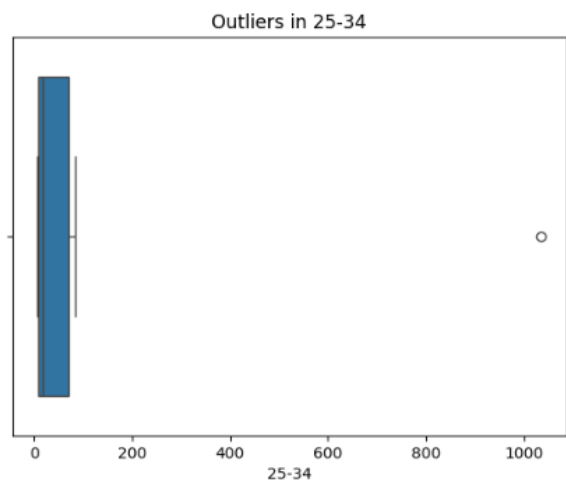
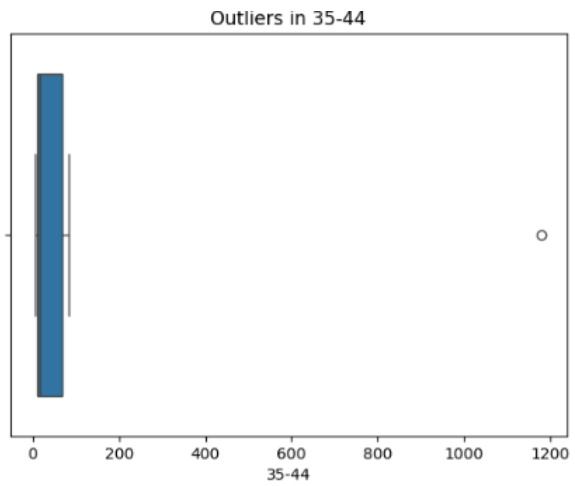
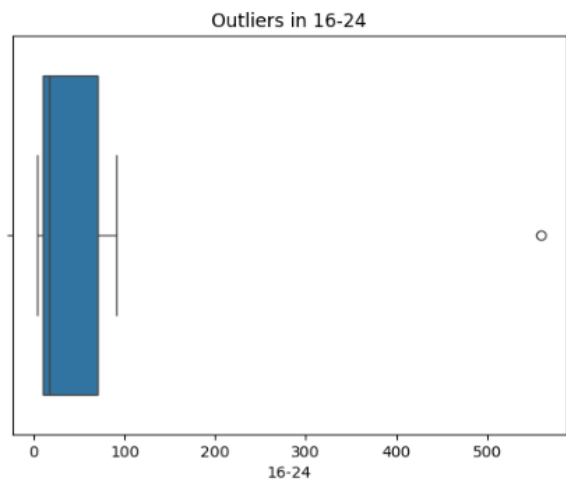
4. Outlier and Variability Analysis

The dataset showed substantial variability in prevalence values across different categories and demographic groups.

Several unusually large values were identified during statistical analysis. These extreme values may reflect:

- Genuine high-prevalence observations
- Data collection variability
- Possible scaling inconsistencies

Clinically, this suggests that mental health burden may not be evenly distributed across all groups and requires careful interpretation.



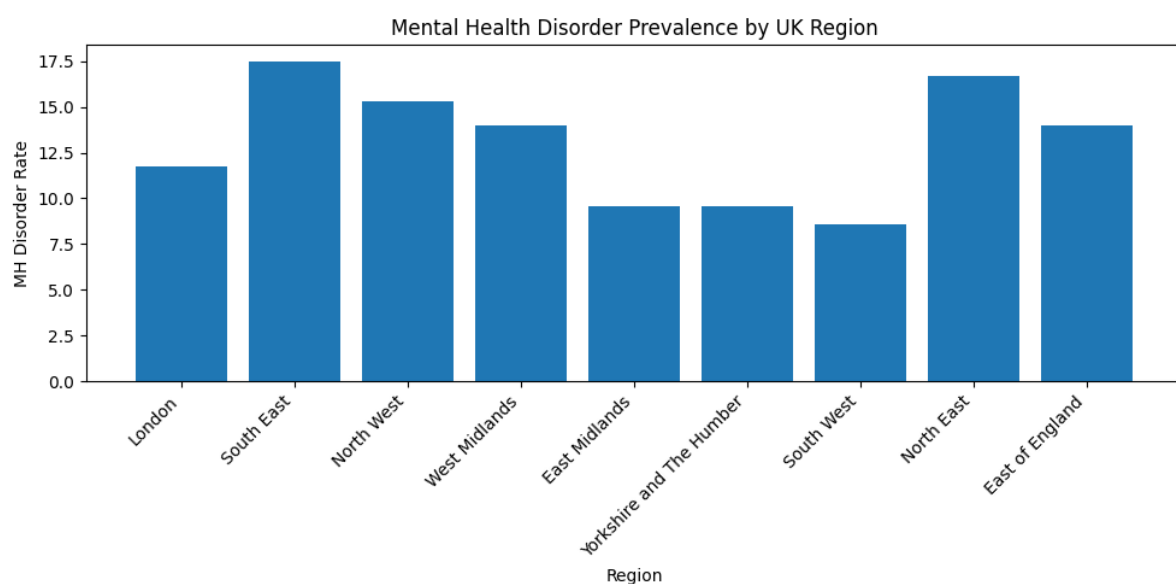
5. Regional Mapping and Visualization

An attempt was made to create regional prevalence maps using choropleth visualization techniques. However, the original dataset did not contain geographic variables such as region or location identifiers.

To demonstrate the mapping methodology:

- A. Synthetic regional labels were temporarily added
- B. Choropleth visualizations were created for demonstration purposes only

This exercise demonstrated how regional mental health patterns could be visualized if real geographic data becomes available in future datasets.



6. Key Clinical Insights

Several practical insights emerged from the Week 1 analysis:

- A. Mental health prevalence varies across demographic groups, particularly by age and gender.
- B. Data completeness remains an important issue, with moderate levels of missing information observed.
- C. Visualization techniques such as grouped bar charts and choropleth maps can support easier interpretation of mental health trends.
- D. Further analysis would benefit from additional geographic and socioeconomic variables.

7. Recommendations for Future Work

Based on the Week 1 findings, the following next steps are recommended:

- A. Incorporate geographic variables to enable true regional analysis.
- B. Include ethnicity and socioeconomic information for deeper demographic insights.
- C. Apply advanced statistical and machine learning methods to identify risk patterns.
- D. Continue improving data quality and completeness before predictive modelling.

Conclusion

Week 1 analysis successfully established a foundational understanding of the APMS-2014 dataset and demonstrated several useful analytical techniques for mental health research. The findings highlight demographic variation in mental health disorder prevalence and emphasize the importance of clean, well-structured data for reliable clinical insights.

Although the dataset had some limitations, particularly the lack of geographic information, the analysis provides a strong starting point for future mental health analytics and decision-support applications within NHS-related research contexts.